

Monitoring the growth of American beech affected by beech bark disease in Maine using the Kalman filter

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The effect of beech bark disease on tree growth was tracked using paired resistant and susceptible American beech trees in two locations in Maine. Within each site, the paired trees were chosen in close proximity and with similar morphological characteristics (e.g. stem diameter and crown class) to minimize environment effects in subsequent analysis. A Kalman filter approach was employed to analyse the yearly time-dependent mean differences between paired susceptible and resistant tree-ring widths using simple structural time series models in state space form. On one site, under the influence of a moderate maritime climate, stand dynamics is hypothesized to account for the 34 year difference in onset of decline of trees in codominant, versus those in the intermediate crown classes. The harsher winter conditions associated with the second, more northerly site and known to limit the insect component of the disease complex, are hypothesized to be more of a factor in the close (six year) difference in decline onset between the two crown classes on this site. Some strengths and cautions in the Kalman filter approach are discussed in relation to the analysis of time-dependent trends in tree-ring series.

Keywords: Dendrochronology; maximum likelihood estimation; structural time series; tree-ring analysis

1. Introduction

Beech bark disease (BBD) is a major dieback–decline disease that causes significant mortality and defect in American beech *Fagus grandifolia* Ehrh. It results when bark, having been attacked and altered by the beech scale *Cryptococcus fagisuga* Lind., is invaded and killed by fungi, primarily *Nectria coccinea* var. *faginata* Lohman, Watson and Ayers, and sometimes *Nectria galligena* Bres. (Ehrlich, 1934; Lohman and Watson, 1943; Cotter and Blanchard, 1981). Sometime before 1890, the scale was accidentally brought to Halifax, Nova Scotia on nursery stock of European beech (*Fagus sylvatica* L.); by 1925, it was present throughout mainland Nova Scotia (Ehrlich, 1934; Hawboldt, 1944). By 1932, the scale and associated *Nectria* fungus were killing trees throughout the mature beech stands of the Maritime Provinces and in localized areas of eastern and south-central Maine. The scale insect has continued to spread, establishing its current range in the north into Quebec, west to Ohio, and south to Tennessee (Houston, 1994).

The pattern of insect spread and subsequent occurrence of *Nectria* infection and tree death or defect has led to an arbitrary classification of disease development over time and space (Shigo, 1972). In the ‘advancing front’, areas recently invaded by the beech scale are characterized by forests

with many large, old trees supporting scattered, sparse, building populations of the insect. In the 'killing front', areas are characterized by high populations of beech scale, severe *Nectria* attacks, and heavy tree mortality. The 'aftermath zone', consists of areas where heavy tree mortality occurred at some time in the past and which now possess some residual big trees and many stands of small trees of both seedling and root sprout origin. Often, these stands are rendered highly defective through the interactions of established populations of beech scale, *Nectria* fungi, and another scale insect, *Xylococcus betulae* (Perg.) Morrison, though widespread mortality due to the disease complex is rare (Houston, 1975).

Some trees remain disease-free in the aftermath stands. Challenge trials have shown these trees to be resistant to the beech scale (Houston, 1982, 1983). Resistant beech trees are found in relatively low numbers, however. In two stands in Nova Scotia, for example, fewer than one per cent of the trees were free of insects or disease (Houston, 1983).

Several studies have been conducted on the growth and mortality trends in stands affected for different lengths of time. Mize and Lea (1979) used *t*-tests on remeasurement data to gauge mortality and decline of individual beech trees in a northern hardwood stand in the Adirondack mountains of New York. According to their data, mortality associated with the killing front, restricted mostly to trees of larger diameter at breast height (dbh) and crown size, began in 1967, leaving an aftermath type forest from 1974 onwards. Twery and Patterson (1984), using remeasurement data on stands in western Massachusetts and northern New Hampshire, found disease-caused mortality in beech to be correlated with stand species composition—those stands dominated by hemlock were the ones with greatest beech mortality. On another stand in the Adirondacks, Runkle (1990) used remeasurement data to find a 25 per cent decrease in beech basal area over the eight-year period 1978–86.

Houston and Valentine (1988) dissected individual beech trees in two aftermath stands in north-eastern Maine to assess the development of defect caused by BBD over time in relation to climate. Regression analysis indicated that canker area increased with time and that current year's canker area was related to both fall precipitation level and winter temperature severity in the previous year. Gavin and Peart (1993) were also interested in defect severity in their analysis of two aftermath stands in northern New Hampshire. Individual trees were rated by defect severity indices and grouped according to three broad levels of defect. Using mean ring-width chronologies developed from tree sections and increment cores, adjusted for the effects of tree size, they found significant differences between the more severely infected trees versus those determined to be uninfected. Evidently, the adjusted individual yearly ring-widths were considered independent for each chronology and *t*-tests were applied, as in Mize and Lea (1979), to judge the significance of changes in growth trends, though their procedure is not explicitly stated.

Tree-ring data are universally recognized as falling under the purview of time series data because of the serial autocorrelation structure inherent in tree growth through a process known as physiological preconditioning (Fritts, 1976, p. 193); many studies have exploited this characteristic with great benefit. Numerous statistical time series techniques have been employed in the analysis of tree-ring data including ARIMA and Kalman filter models. However, the modelling of temporal trends effectively eliminates those methods, like ARIMA models, that are based on the stationarity assumption. Application of the Kalman filter to time series data, however, does not rely on the hypothesis of stationarity. Indeed, several studies have used the Kalman filter for general time series forecasting (Harvey, 1984), trend estimation in tree-ring series (Visser and Molenaar, 1990, 1992), model-based tree-ring analysis of growth trends (Van Deusen, 1989, 1990a), and trend monitoring of paired diseased/resistant tree-rings (Van Deusen, 1991a). In each case, undifferenced time series were used to preserve the trends that were to be revealed as part of the analysis.

The main objectives of our study were to (*i*) determine the long-term growth trend difference

between trees that are susceptible to BBD and those that are resistant to the disease, by crown class; and (ii) determine the timing of significant growth reduction in the susceptible trees relative to their paired resistant counterparts for trees on our two study areas. A Kalman filter modelling approach to trend monitoring of tree-ring data is presented to accomplish these objectives.

2. Methods

2.1 Study areas

Two study sites were sampled in Maine as shown in Fig. 1; the inland study area is termed the 'Northern' site while the area closest to the coast is termed the 'Downeast' site. The Downeast site is comprised of three, 10 ha stands located within 20 km of each other on lands of Champion International Paper Company. All resistant trees in these stands with dbh ≥ 20 cm (dbh is measured at a height of 1.37 m) were located as part of a larger study of the genetic relationships of resistant trees. All three stands were located on gentle slopes (less than 10 per cent) at elevations ranging from approximately 100 to 200 m. The major species by per cent stems ha⁻¹ were American beech (62 to 80 per cent) and sugar maple (*Acer saccharum* Marsh., 12 to 14 per cent). Minor species occurring on the three stands were yellow birch (*Betula aleghaniensis* Britton), white ash (*Fraxinus americana* L.), and eastern hemlock (*Tsuga canadensis* (L.) Carr.). Stand two also had a moderate component (8 per cent) of red spruce (*Picea rubens* Sarg.) that did not occur on the other two stands.

The Northern site comprised an area of 60 ha on the State of Maine Public Land near Seboeis. The site straddles a broad north-south running ridge with portions sloping both to the east and west with maximum elevation of approximately 230 m and an average of 10 per cent slope. Trees in this site were located during the course of a larger study comparing the root-sprouting response of resistant and susceptible trees. As in the Downeast site, only trees with dbh ≥ 20 cm were used. Major tree species were American beech (75 per cent), sugar maple (11 per cent) and yellow birch (7 per cent), with red spruce and white ash comprising the remainder.

On both sites increment cores taken parallel to the slope contour were collected from trees at dbh. The increment cores were prepared and read on a Digital Positioner by the US Forest Service Tree Ring Laboratory in Durham, New Hampshire. Individual tree ring-widths—the width of the individual annual growth increments along the radius determined by the increment core—were measured to the nearest 0.01 mm for each year present on each core. Trees were also classified by crown class according to the following guidelines (Smith, 1962): *dominant* crowns extend above the general level of main crown canopy and receive full sunlight from above and some from the sides; *codominant* crowns form the main crown canopy and receive full light from above, but little from the sides; *intermediate* crowns extend into the main canopy, but receive little sunlight, and only from directly above; *suppressed* crowns are entirely below the level of the main crown canopy. In addition, all beech trees with dbh ≥ 20 cm determined to be free from beech scale attack and/or any sign of defect associated with the BBD complex, were located and mapped. A random sample of these resistant trees were chosen to be paired with the nearest diseased tree of equal crown class and dbh. The susceptible trees used in the pairing possessed crowns that showed evidence of decline caused by BBD; such evidence normally manifests itself by trees having thinned crowns, small chlorotic leaves, and branch dieback.

2.2 Structural time series models

Univariate structural time series models provide a medium that allows the incorporation of a simple

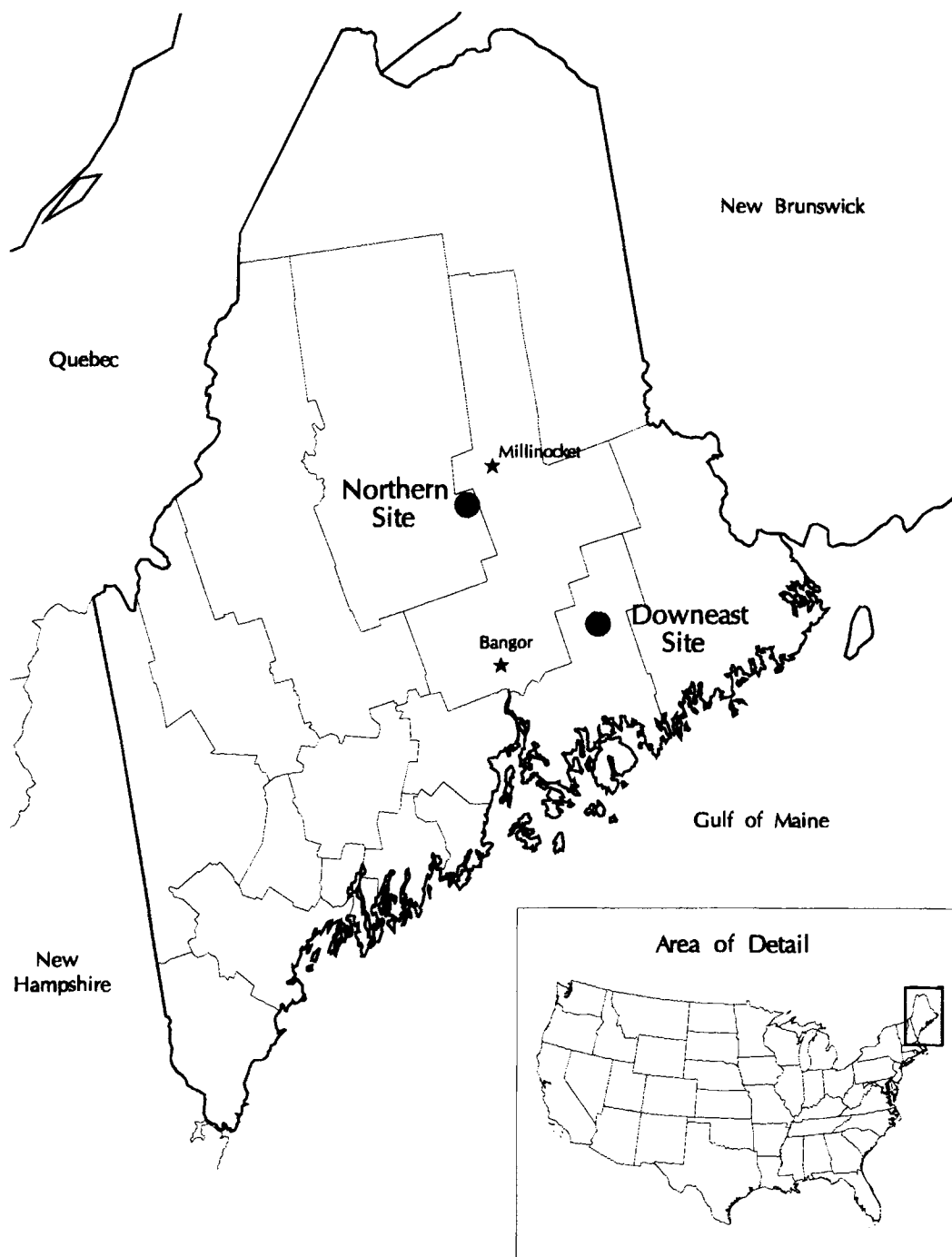


Figure 1. Location of the study areas in Maine.

predictive model into the analysis of a time series. The predictive model is meant to be descriptive of the time series, and is composed of estimable components which have a direct interpretation. Two structural models that are commonly used in econometrics (Harvey, 1989) are employed here for the beech data—both allow for the modelling of trend (i.e., long-term changes) in the series.

The first and simplest structural model used in this study is the random walk plus noise (RWPN) model. The RWPN model views the time series observations (y_t) as being composed of a cumulative level component (μ_t) plus a disturbance term (ξ_t). The cumulative level component is generated by a simple random walk model. Neither the cumulative level, nor the disturbance terms are directly observable. The RWPN model is given as

$$y_t = \mu_t + \xi_t \quad \xi_t \sim \mathcal{N}(0, h_t)$$

$$\mu_t = \mu_{t-1} + \eta_t \quad \eta_t \sim \mathcal{N}(0, q_{\mu t})$$

The two disturbance terms are normally distributed, mutually and serially independent random variables with zero mean. The variance of these two terms may be set up as either being constant or nonconstant as will be described later. Inasmuch as μ_t follows a random walk, this model allows for variation in the level of the process; if the disturbance variance $q_{\mu t} \equiv 0$, then the level of the process is assumed constant and the observations may best be estimated with the sample mean. To our knowledge, the RWPN model has not been previously employed in tree-ring studies; its reduced form is equivalent to an ARIMA(0,1,1) model (Harvey, 1989, p. 68).

The second structural model is known as the local linear trend (LLT) model. The LLT model has been used successfully in previous tree-ring studies by Visser and Molenaar (1990) and Van Deusen (1991a, 1991b). The LLT model may be written as the following set of equations

$$y_t = \mu_t + \xi_t \quad \xi_t \sim \mathcal{N}(0, h_t)$$

$$\mu_t = \mu_{t-1} + \beta_{t-1} + \eta_{\mu t} \quad \eta_{\mu t} \sim \mathcal{N}(0, q_{\mu t})$$

$$\beta_t = \beta_{t-1} + \eta_{\beta t} \quad \eta_{\beta t} \sim \mathcal{N}(0, q_{\beta t})$$

The LLT model adds to the RWPN model a stochastic slope component which follows a random walk. As in the RWPN model, all three of the disturbance terms are assumed to be mutually and serially independent. The disturbance terms $\eta_{\mu t}$ and $\eta_{\beta t}$ allow the level and slope of the trend to change. The reduced form of the LLT model is equivalent to an ARIMA(0,2,2) model (Harvey, 1989, p. 68).

The RWPN and LLT models are related through an intermediate model with constant slope known as the random walk plus drift (RWPd). The RWPd model proceeds directly from the LLT model by assuming that $q_{\beta t} \equiv 0$; the slope parameter, β , therefore becomes a constant component in this intermediate model. Applying the further assumption that $\beta \equiv 0$ to the RWPd formulation yields the RWPN model (Harvey, 1989, p. 38).

2.3 State space model formulation

The key to implementing any type of time series model, structural or otherwise, with the Kalman filter, is to be able to write the model in state space form (SSF). Harvey (1989, p. 100) gives the general SSF for multivariate and univariate time series. Following Harvey, the SSF for univariate time series to be used in this paper is given by the following two equations

$$y_t = \mathbf{z}'_t \boldsymbol{\alpha}_t + \xi_t \tag{1a}$$

$$\boldsymbol{\alpha}_t = \mathbf{T}_t \boldsymbol{\alpha}_{t-1} + \boldsymbol{\eta}_t \tag{1b}$$

Table 1. Linkages between the structural time series models and the state space model formulation

Structural model	State space model component					
	p	$\underline{\alpha}_t$	$\underline{z}_t$	$\mathbf{T}_t$	$\underline{\eta}_t$	$\mathbf{Q}_t$
RWPN	1	$[\mu_t]$	$[1]$	$[1]$	$[\eta_t]$	$[q_{\mu t}]$
LLTM	2	$\begin{bmatrix} \mu_t \\ \beta_t \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \eta_{\mu t} \\ \eta_{\beta t} \end{bmatrix}$	$\begin{bmatrix} q_{\mu t} & 0 \\ 0 & q_{\beta t} \end{bmatrix}$

known as the *measurement* and *transition* equations, respectively. The p model parameters, or state variables, in these SSF equations are determined by the structural time series model used; y_t is the univariate time series of interest, $\underline{z}_t$ ($p \times 1$) and $\mathbf{T}_t$ ($p \times p$) are both system matrices, and $\underline{\alpha}_t$ ($p \times 1$) is the state vector following the first-order Markov process (1b). The disturbance terms ξ_t and $\underline{\eta}_t$ ($p \times 1$) are assumed to be serially independent and normally distributed such that $\xi_t \sim \mathcal{N}(0, h_t)$ and $\underline{\eta}_t \sim \mathcal{N}(0, \mathbf{Q}_t)$. In addition, ξ_t and $\underline{\eta}_t$ are assumed to be independent of one another and the initial state ($\underline{\alpha}_0$); that is

$$\begin{bmatrix} \xi_t \\ \underline{\eta}_t \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} h_t & 0' \\ 0 & \mathbf{Q}_t \end{bmatrix} \right) \quad (2)$$

and

$$E[\xi_t \underline{\alpha}_0'] = 0' \quad E[\underline{\eta}_t \underline{\alpha}_0'] = \mathbf{0} \quad t = 1, \dots, T$$

where T is the length of the time series. The disturbance terms and the initial state vector are assumed to be normally distributed so that the Kalman filter will be an optimal estimator of the state vector—optimal in the sense that it yields the conditional mean (conditional on all information available at time t) as the minimum mean square error estimator of $\underline{\alpha}_t$ (Harvey, 1989, p. 105; Maybeck, 1979, p. 232).

The structural time series models presented above may easily be written in SSF. Inspection of the RWPN model indicates that there is only one state variable (μ_t) so that $\underline{\alpha}_t$ becomes a scalar in the SSF. Therefore, the transition equation becomes a scalar equation as does all its associated components.

The LLT model is slightly more complicated. It has two state variables (μ_t and β_t) comprising the state vector with the other matrices conformably dimensioned. Table 1 presents the SSF matrices for both structural time series models using the components of each model where appropriate. Substitution of the matrices in Table 1 associated with each structural time series model into equation set (1) provides the link with the SSF formulation.

2.4 The Kalman filter

The Kalman filter is composed of two sets of equations that are applied recursively to the time series. The first set of equations is applied at time $t - 1$ to *predict* a new state at time t ; subsequently, when the data at time t become available, the predicted estimate is *updated*.

The connection between the SSF of the time series model and the Kalman filter recursive equations may be seen by deriving these equations using multivariate normal distribution theory and a Bayesian argument. A sketch of this derivation is provided here to illustrate the link; Harvey (1989, p. 109) and Meinhold and Singpurwalla (1983) provide more details.

The assumption of multivariate normality of the error terms in the SSF is set forth in (2). In addition to this, it is assumed that the initial state vector at time $t = 0$ also has a multivariate normal distribution given by $\underline{\alpha}_0 \sim \mathcal{N}_p(\underline{a}_0, \mathbf{P}_0)$. Now, by (1b), the state vector at time $t = 1$ is given as $\underline{\alpha}_1 = \mathbf{T}_1 \underline{\alpha}_0 + \underline{\eta}_1$. Taking the expectation and variance of $\underline{\alpha}_1$ yields

$$\underline{a}_{1|0} = E[\underline{\alpha}_1] = \mathbf{T}_1 \underline{a}_0 \quad (3a)$$

$$\mathbf{P}_{1|0} = \text{Var}(\underline{\alpha}_1) = \mathbf{T}_1 \mathbf{P}_0 \mathbf{T}_1' + \mathbf{Q}_1 \quad (3b)$$

From equations (3) it may be seen that the *prior* distribution for the state vector at time $t = 1$ is multivariate normal $\underline{\alpha}_1 \sim \mathcal{N}_p(\underline{a}_{1|0}, \mathbf{P}_{1|0})$. The equations in (3) are known as the Kalman filter *prediction* equations.

Once an observation (y_1) has become available at time $t = 1$, the distribution of $\underline{\alpha}_1$ conditional on y_1 is required. To get this distribution, the joint distribution of $[\underline{\alpha}_1, y_1]$ is required. This may be derived after determining the distribution of y_1 using the measurement equation from the SSF: $y_1 = \underline{z}_1' \underline{\alpha}_1 + \xi_1$. As before, taking the expectation and the variance yields a normal distribution given by $y_1 \sim \mathcal{N}(\underline{z}_1' \underline{a}_{1|0}, \underline{z}_1' \mathbf{P}_{1|0} \underline{z}_1 + h_1)$. The covariance may be derived as $\text{Cov}(\underline{\alpha}_1, y_1) = \underline{z}_1' \mathbf{P}_{1|0}$. Therefore, the joint distribution of the predicted state vector and observation is given as

$$\begin{bmatrix} \underline{\alpha}_1 \\ y_1 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \underline{a}_{1|0} \\ \underline{z}_1' \underline{a}_{1|0} \end{bmatrix}, \begin{bmatrix} \mathbf{P}_{1|0} & \mathbf{P}_{1|0} \underline{z}_1 \\ \underline{z}_1' \mathbf{P}_{1|0} & \underline{z}_1' \mathbf{P}_{1|0} \underline{z}_1 + h_1 \end{bmatrix} \right) \quad (4)$$

Applying the well-known lemma on obtaining the conditional distribution from a multivariate normal distribution (Harvey, 1989, p. 165) yields the *posterior* distribution of $\underline{\alpha}_1$, conditional on the value of y_1 . This distribution is multivariate normal with mean and variance given as

$$\underline{a}_1 = \underline{a}_{1|0} + \mathbf{P}_{1|0} \underline{z}_1 f_1^{-1} (y_1 - \underline{z}_1' \underline{a}_{1|0}) \quad (5a)$$

$$\mathbf{P}_1 = \mathbf{P}_{1|0} - \mathbf{P}_{1|0} \underline{z}_1 f_1^{-1} \underline{z}_1' \mathbf{P}_{1|0} \quad (5b)$$

where

$$f_1 = \underline{z}_1' \mathbf{P}_{1|0} \underline{z}_1 + h_1$$

Equations (5) make up the Kalman filter *update* equations. Both the prediction and update equations take this same form over time by generalizing the time subscripts to t and $t - 1$ in (3) and (5), where $t = 1, \dots, T$.

Though the form of the update equations in (5) may be used in the Kalman filter recursions, a more computationally stable form is often used (Grewal and Andrews, 1993, p. 113). For a given time t , first define

$$\nu_t = y_t - \underline{z}_t' \underline{a}_{t|t-1}$$

$$\underline{k}_t = \mathbf{P}_{t|t-1} \underline{z}_t f_t^{-1}$$

where ν_t is the *prediction error* or *innovation* and $\underline{k}_t$ is the Kalman *gain* matrix. Upon substitution of these quantities into (5) and some algebra, the update equations take the form

$$\underline{a}_t = \underline{a}_{t|t-1} + \underline{k}_t \nu_t \quad (6a)$$

$$\mathbf{P}_t = [\mathbf{I} - \underline{k}_t \underline{z}_t'] \mathbf{P}_{t|t-1} [\mathbf{I} - \underline{k}_t \underline{z}_t']' + \underline{k}_t h_t \underline{k}_t' \quad (6b)$$

The new form for the update covariance matrix (6b) is known as the 'Joseph form'. To implement the Kalman filter recursion equations given in (3) and (6), values for the state vector and update covariance matrix at the initial time period are required (refer to (3)). Normally, values of $\underline{a}_0 = \underline{0}$ and $\mathbf{P}_0 = \kappa \mathbf{I}$ are used, where κ is chosen to be very large so that $\mathbf{P}_0^{-1} = \mathbf{0}$ (Harvey, 1989, p. 121) and $\mathbf{I}$ is an identity matrix of dimension $p \times p$.

2.5 Maximum likelihood estimation (MLE)

The two structural time series models described have unknown parameters that do not belong to the state vector ($\underline{\alpha}_t$), but which must be estimated. In both models these unknown parameters, termed *hyperparameters*, belong to the variance components of the disturbance terms (h_t and $\mathbf{Q}_t$) which, in turn, govern the stochastic progression of the state variables. These hyperparameters may all be collected into an $m \times 1$ vector $\underline{\psi}$ which is estimated by the method of maximum likelihood, subject to the conditions that $\underline{\psi} > \underline{0}_{(m \times 1)}$. Harvey (1989) and Engle and Watson (1981) give the details of MLE of the hyperparameters and generalizations to the multivariate time series case. In brief, the *prediction error decomposition* form of the log-likelihood for any univariate state space model may be written in terms of the innovations as

$$\mathcal{L}(\underline{\psi}) = \log L(\underline{\psi}) = \sum_{t=d+1}^T \left\{ -\frac{1}{2} \log 2\pi - \frac{1}{2} \log f_t - \frac{1}{2} \nu_t^2 f_t^{-1} \right\}$$

Note that the first d time intervals are skipped in the summation because it takes a few iterations for the Kalman filter covariance matrices to stabilize after the initial settings. In this study several values for d were tried and $d = p + 1$ was found to be the most reasonable.

Two methods of optimization were used in this study to maximize the log-likelihood. The first is the derivative-free simplex method with line search; interested readers may consult Press *et al.* (1986, p. 289) for algorithmic details and source listings. The method of *scoring* was the second method used. It is based on the first derivatives of $\mathcal{L}(\underline{\psi})$ only and uses the score vector and the information matrix. The i th element in the score vector ($\underline{s}(\underline{\psi})$) and the ij th element of the information matrix ($\mathcal{I}(\underline{\psi})$) for univariate time series are given by

$$s_i(\underline{\psi}) = -\frac{1}{2} \sum_{t=1}^T \left\{ \left(f_t^{-1} \frac{\partial f_t}{\partial \psi_i} \right) (1 - f_t^{-1} \nu_t^2) \right\} - \sum_{t=1}^T \left\{ \frac{\partial \nu_t}{\partial \psi_i} f_t^{-1} \nu_t \right\} \quad i = 1, \dots, m$$

and

$$\mathcal{I}_{ij}(\underline{\psi}) = \frac{1}{2} \sum_{t=1}^T \left(f_t^{-2} \frac{\partial f_t}{\partial \psi_i} \frac{\partial f_t}{\partial \psi_j} \right) + \sum_{t=1}^T \left(f_t^{-1} \frac{\partial \nu_t}{\partial \psi_i} \frac{\partial \nu_t}{\partial \psi_j} \right) \quad i, j = 1, \dots, m$$

respectively, where $\partial/\partial \psi_i$ denotes the partial derivative with respect to the i th component of $\underline{\psi}$. Closed-form solutions to the partial derivative terms are given for the general multivariate time series case in Harvey (1989, p. 143).

The method of scoring (Harvey, 1990, p. 134) proceeds by using an iterative update of the hyperparameters from step k to $k + 1$ of the form: $\underline{\psi}^{(k+1)} = \underline{\psi}^{(k)} + \lambda \mathcal{I}^{-1}(\underline{\psi}^{(k)}) \underline{s}(\underline{\psi}^{(k)})$, where $0 < \lambda \leq 1$ is a step length which may be determined by a line search method—a modified golden section search was used here (Press *et al.*, 1986, p. 277). The step length modifier, λ , was found to be very useful in circumstances where one of the hyperparameters was on the boundary of the parameter space; this equates to the associated variance component being zero. Without it, the offending hyperparameter sometimes would go negative, producing a negative variance. A Lagrange multiplier approach

might also have been used to constrain the hyperparameters to be non-negative as an alternative to this method. The entire Kalman filter iteration from time $t = 1, \dots, T$ is run for each step of the scoring algorithm. Scoring converges when the score vector is effectively 0.

2.6 Tree-ring analysis methods

Data from the paired-tree American beech study described earlier were used with the structural time series models and the Kalman filter described in the previous sections to monitor the growth response of diseased and resistant trees. The raw ring-widths were used from each pair of tree cores. Beginning at 1900, for every year that both trees in a pair shared in common, the resistant tree's growth was subtracted from the susceptible's, forming a paired difference ring-width series

$$y_{it} = y_{it}(s) - y_{it}(r) \quad t = 1, \dots, T \quad (7)$$

where $y_{it}(s)$ and $y_{it}(r)$ are the raw ring-widths in year t for the susceptible and resistant tree, respectively, in pair i . Missing values were assigned to those years in the beginning of the series where neither tree, or only one tree in the pair had a growth measurement. Missing years at the beginning of the paired difference ring-width series may have been due either to age differences between the two trees comprising a pair, or to cores that were difficult to measure due to breakage or similar problems.

The tree pairs from the two regions in Maine were analysed separately. Within both regions, each of the paired difference ring-width series (y_{it}) were grouped together into two broad crown class categories based on the crown class designation of the individual trees in the pairs. The two broad crown class categories recognized were termed codominant and intermediate. The codominant group consisted of pairs where both trees were classified as having either dominant or codominant crown class. Such trees are effectively in the the upper part of the main forest canopy, receiving most of the direct solar radiation. The intermediate class contained pairs with at least one tree of intermediate crown class, the other member may be codominant, intermediate, or suppressed. These trees are found in the middle and lower portions of the main canopy and receive a large proportion of shading from the surrounding codominants.

Within each regional crown class category, the pairs (7) were averaged by year to form the univariate time series (y_t) used in the analysis

$$y_t = \sum_{i=1}^{n_t} \frac{y_{it}}{n_t} \quad t = 1, \dots, T \quad (8)$$

The n_t reflects the number of sample pairs available for the mean paired difference series in year t . The sample size is variable because not all pairs shared a common point of origin for the same reasons that individual trees comprising each pair may not have shared a common beginning year. Thus n_t may change over time to accomodate more pairs as yearly measurements become available. This procedure has been successfully used by Van Deusen (1988, 1991a,b). The beginning point for y_t was set circa 1900 to allow enough time for the filter covariances to stabilize and to get a fair estimate of the growth trends prior to the onset of the disease complex. The first year in the time series where $n_t \geq 2$, becomes the actual initial year in y_t . This is required so that between-pair variances can be calculated within each successive year. All pairs within a region shared a common endpoint since all plots in the region were sampled in the same year.

Both the RWPN and LLT models require variance estimates for each of the m disturbance terms in the model. Hence, the RWPN model requires two variance parameters to be estimated and the LLT model requires three. Visser and Molenaar (1988, 1990, 1992), in the majority of cases, estimate

all variance parameters entirely by ML. In some instances, however, they choose to set values for the parameters by hand. Van Deusen (1988, 1990b, 1991a,b) breaks each variance component into two parts. The first part may be calculated from the within-year measurements directly for all cores in the chronology; the second part is estimated by ML.

In addition to the different variance structures described above, several studies (Visser and Molenaar 1988, 1990, 1992; Van Deusen 1991a,b) have used the concentrated form of the log-likelihood that assumes an underlying variance component that is common to both h_t and Q_t . This variance component is concentrated out of the log-likelihood and estimated in closed-form, thereby simplifying the log-likelihood and associated optimization through hyperparameter space reduction. We prefer the unconcentrated form given by $\mathcal{L}(\psi)$ because it allows more flexibility for independent variance components in each model. In addition, we estimate each variance component in two parts following the models of Van Deusen (1988, 1990b, 1991a,b).

Two different variance models were tried for both the RWP and the LLT model; these took the form of time-variant and time-invariant variance models. For the time-variant variance model, the $m = 2$ variance components in the RWP model were defined as $h_t = \sigma_{yt}^2 \psi_y$ and $Q_t = q_{\mu t} = \sigma_{yt}^2 \psi_\mu$, a scalar for the RWP. The hyperparameters $\underline{\psi} = [\psi_y \ \psi_\mu]'$ are estimated by ML, while σ_{yt}^2 is calculated as the variance of the mean paired growth difference (Van Deusen, 1991a)

$$\sigma_{yt}^2 = \frac{\sum_{i=1}^{n_t} (y_{it} - \bar{y}_t)^2}{n_t(n_t - 1)} \quad t = 1, \dots, T$$

This form of variance calculation allows different variances for each year in the time series because the components of $\underline{\psi}$ estimated by ML are constant over time. Two of the $m = 3$ variance components for the LLT model are calculated in the same manner as in the RWP model. The variance components for the LLT model are given as

$$h_t = \sigma_{yt}^2 \psi_y$$

and

$$Q_t = \begin{bmatrix} q_{\mu t} & 0 \\ 0 & q_{\beta t} \end{bmatrix} = \begin{bmatrix} \sigma_{yt}^2 \psi_\mu & 0 \\ 0 & \sigma_{\beta t}^2 \psi_\beta \end{bmatrix}$$

All three components of $\underline{\psi} = [\psi_y \ \psi_\mu \ \psi_\beta]'$ are estimated by ML. The final component is calculated as the variance of the mean slope

$$\sigma_{\beta t}^2 = \frac{\sum_{i=1}^{n_t} (b_{it} - \bar{b}_t)^2}{n_t(n_t - 1)} \quad t = 2, \dots, T$$

This variance is also time-variant, changing yearly. The yearly slope (change in growth) for each tree pair is calculated as

$$b_{it} = \Delta y_{it} = y_{it} - y_{i(t-1)}$$

where Δ is the first difference operator. Then b_t simply becomes the average slope for all pairs in year t given by

$$b_t = \sum_{i=1}^{n_t} \frac{b_{it}}{n_t} \quad t = 2, \dots, T$$

One year is lost in the series because of the lag calculation in this operation. The slope at time t may also be calculated by averaging the first differences from consecutive years on either side of t ;

however, two years are lost from the series and future data is used in this calculation—using the next year's growth in the estimation of the current year's growth seems biologically indefensible.

The time-invariant variance models are defined similarly for both structural time series models used. These time-invariant models simply use the average variance over all years; that is

$$\sigma_y^2 = \sum_{t=1}^T \frac{\sigma_{y_t}^2}{T} \quad \text{and} \quad \sigma_\beta^2 = \sum_{t=2}^T \frac{\sigma_{\beta_t}^2}{T-1}$$

are substituted into the components h_t and $\mathbf{Q}_t$ in place of $\sigma_{y_t}^2$ and $\sigma_{\beta_t}^2$.

Combining the two different structural time series models (RWPN and LLT) with the two different disturbance variance models (time-variant and time-invariant) yielded a total of four different models that were fitted to each of the regional crown class mean paired difference series.

2.7 Model diagnostics

The innovations ν_t contain the new information on the latest observation y_t available after a prediction step. From the preceding distributional arguments on the derivation of the Kalman filter it can be shown that the innovations are independent and normally distributed such that $\nu_t \sim \mathcal{N}(0, f_t)$. In addition, since there is no requirement for constant variance, we may define the *standardized prediction errors* $\tilde{\nu}_t = \nu_t / \sqrt{f_t} \sim \mathcal{N}(0, 1)$ to use in evaluation of misspecification of the residuals. These results hold approximately when parameters have been estimated by ML (Harvey, 1989, p. 256). Four simple diagnostic criteria were used in evaluating the standardized residuals for each of the four different models fitted; these were:

- (i) The sample autocorrelation function for the first $\zeta = \max[\sqrt{T}, \log(T), T/4]$ autocorrelations ($r_{\tilde{\nu}_t}$) with asymptotic standard error limits (Vandale, 1983, p. 106).
- (ii) Box-Ljung Q statistic for the first ζ autocorrelations, where $Q = (T-d)(T-d+2) \sum_{\tau=1}^{\zeta} (T-d-\tau)^{-1} r_{\tilde{\nu}_t}^2(\tau)$ (Harvey, 1989 p. 259; Vandale, 1983 p. 108).
- (iii) Normal probability plot of the $\tilde{\nu}_t$ with associated correlation coefficient.
- (iv) Standardized residual plots.

In addition to testing the aptness of the $\tilde{\nu}_t$ to the assumption of independence and normality for each model, two other criteria were employed for model selection. The first criterion evaluates the state vector with respect to convergence of the method of scoring to an acceptable optimum. In several cases the LLT model turned out to be numerically inappropriate. In these instances, the variance parameter ψ_β associated with the slope state variable was found to lie on the boundary of the parameter space at $\psi_\beta = 0$. When this happened, its associated derivative component in the score vector would normally have a large negative value. The score component for the other two variance parameters, however, normally were close to zero. When this was the case, the parameter estimates for ψ_y and ψ_μ were quite close to the RWPN model, indicating that the simple parameterization was more appropriate for those data.

The second criterion used for model selection was the Schwartz information criterion (SIC). Koehler and Murphree (1988) and Van Deusen (1990a,b) both prefer SIC over the Akaike information criterion (AIC) because the SIC leads to choosing a lower order model. The rule for the SIC is to choose the model that minimizes $-2\mathcal{L}(\underline{\psi}) + (p+m)\ln(T-d)$ (Van Deusen, 1990a,b). This criterion may be used for comparison among all of the four models. Harvey (1989, p. 102) argues that the model that minimizes the length of the state vector (and thus the length of $\underline{\psi}$ in the structural models considered here) is 'a basic criterion for a good state space representation'. The use of SIC seems consistent with that goal.

Finally, the predictor–corrector structure of the Kalman filter may also be useful in overall model evaluation. The components of $0 \leq k_t \leq 1$ are the optimal weighting factors through which the residual ν_t is passed for a correction to be added to the predicted state (Maybeck, 1979, p. 218). The effect of k_t is most easily seen in the RWPN model where the state variable is scalar. In this model, equation (6a) becomes

$$a_t = a_{t|t-1} + k_t(y_t - a_{t|t-1}) \quad (9)$$

If $k_t = 1$ then $a_t = y_t$; that is, the optimal estimate is simply the newest observation—the optimal prediction from $t - 1$ is completely discounted. Conversely, if $k_t = 0$ then $a_t = a_{t|t-1}$ and the observation at time t is completely ignored. This phenomenon is under the control of the disturbance variance terms which are partially estimated by ML. In general, if Q_t is large relative to h_t , then there is more uncertainty in the adequacy of the structural time series model to portray the dynamics of the series accurately. In this case, the filter gains will tend to increase to weight the measurements more heavily. Similarly, if h_t is large relative to Q_t , this would indicate that the measurements are more prone to noise with larger disturbances, and the filter predictions would dominate (Maybeck, 1979, p. 224).

The Kalman gains should be plotted over time to help with evaluating the model. The usefulness of the gains for model evaluation has been somewhat neglected in the statistical literature, probably because of the lack of a formal inference mechanism. However, their usage in engineering applications is well established.

3. Results

The distributional assumptions associated with the Kalman filter allow the placement of confidence intervals (more precisely termed probability intervals in a Bayesian context) around the optimal estimates of the state vector in each time period $t = d + 1, \dots, T$. While confidence intervals placed around the slope parameter in the LLT model are helpful in determining if the slope is non-zero in any given year, the more important intervals for trend monitoring are associated with the level parameter in both models. Under the posterior normality assumption, the estimated 95 per cent confidence interval endpoints for the level are given by

$$\mu_t \in \underline{a}_t[1] \pm 2 \times \mathbf{P}_t[1, 1] \quad t = d + 1, \dots, T$$

where the brackets denote subscripts into the update state vector and covariance matrix given in (6). Figures 2 and 4 present the time series plots of y_t with the Kalman filter estimate and upper and lower confidence intervals for $t = d + 1, \dots, T$. In these plots, the years where the upper and lower confidence intervals include zero have no significant difference in growth between resistant and susceptible trees. Years where both confidence interval endpoints lie above the zero line are those where the susceptible trees are growing significantly faster than the resistants; similarly, when the resistant trees are growing significantly better than the susceptible trees, both confidence intervals will lie below the zero line. This analysis method allows the interpretation of the graphs for significant trends in growth differences between the sets of resistant and susceptible trees, and is used in the following interpretations.

3.1 Downeast site

The observed mean paired difference time series for both the codominant and intermediate crown classes on the Downeast site had a total length of $T = 88$ years, beginning in 1900 and ending in the

year of measurement, 1987. The optimal estimate from the time-invariant Kalman filter for the RWPN model was the best fit for the observed codominant time series (Fig. 2a), with the steady state Kalman gain converging to $k_t = 0.776$ after the first four years. The growth of the susceptible trees was significantly better than that of the resistant trees until the late 1930s (Fig. 2a). The trend from that time forward was a decline in growth of susceptible trees relative to their resistant counterparts, which became significant in 1941 and continued thusly except for two years in the mid 1950s (1956 and 1957). The minimum difference in growth estimates occurred in 1986 where $a_t = -1.47$ mm.

The best fitting model for the intermediate trees (Fig. 2b) was also the time-invariant RWPN model with steady state $k_t = 0.695$. Unlike the codominants, the early periods of growth (under suppression of the codominants) showed no significant difference in growth between the susceptible and resistant trees until 1944 when the susceptible trees grew significantly faster than their resistant counterparts. This was followed by a second period of equal growth between 1962 and 1974. In 1975 the susceptible trees began to manifest the same downward growth trend as the codominants; the minimum estimated difference of $a_t = -0.88$ mm occurred in 1986.

In comparing the codominant pairs with the intermediate pairs (Fig. 3), it is interesting to note that the response of the intermediate pairs from circa 1940 on was very similar to that of the codominant pairs from circa 1900. In fact, a visual shifting of the codominant graph 40 years ahead shows that the only real difference in the two trends is in the magnitude of growth difference between susceptible and resistant pairs in the codominants versus the intermediates. This difference is quite reasonable given that the codominants were still present in the stand and competing with the newly released intermediates, albeit in a less pronounced manner because of the loss of vigour in the codominant crowns associated with their decline.

Growth and development of *C. fagisuga* populations are best on the larger more vigorous trees that can provide in abundance, the sugars and amino acids necessary for growth and survival; these trees also provide more of the protective spatial bark habitat features utilized by the insect. Therefore, fast-growing susceptible codominant trees were probably the most severely affected initially. Following significant invasion of scale-altered bark by the *Nectria* fungi, growth of the susceptible codominants began declining (the late 1930s). Many other large trees were killed outright in these stands as the disease epidemic developed, and stands were often opened up significantly. Increased availability of light to the intermediates stimulated the growth rates, especially of the more vigorous (susceptible) trees relative to the less vigorous resistants. The scale infestation/fungus infection process was then repeated in a similar manner on this newly 'recruited' cohort. The time lag between the *significant* decline of the codominants and the subsequent *significant* increase of the intermediates was three years. However, it must be emphasized that the actual time of infection, subsequent crown thinning and associated canopy opening is unknown and may be closer to ten or more years.

3.2 Northern site

The observed mean paired difference time series for the codominant crown class on the Northern site began in 1909 and ended in 1990 for a total of $T = 82$ years; the intermediate crown class series had a length of $T = 80$ years beginning in 1911. The time-variant RWPN model with average $k_t = 0.794$ was the best fit for the codominant pairs. The growth response of these trees (Fig. 4a) was similar to those on the Downeast site. The susceptible trees began by being more vigorous, then grew approximately the same as the resistant trees in the mid 1930s, until they began their final

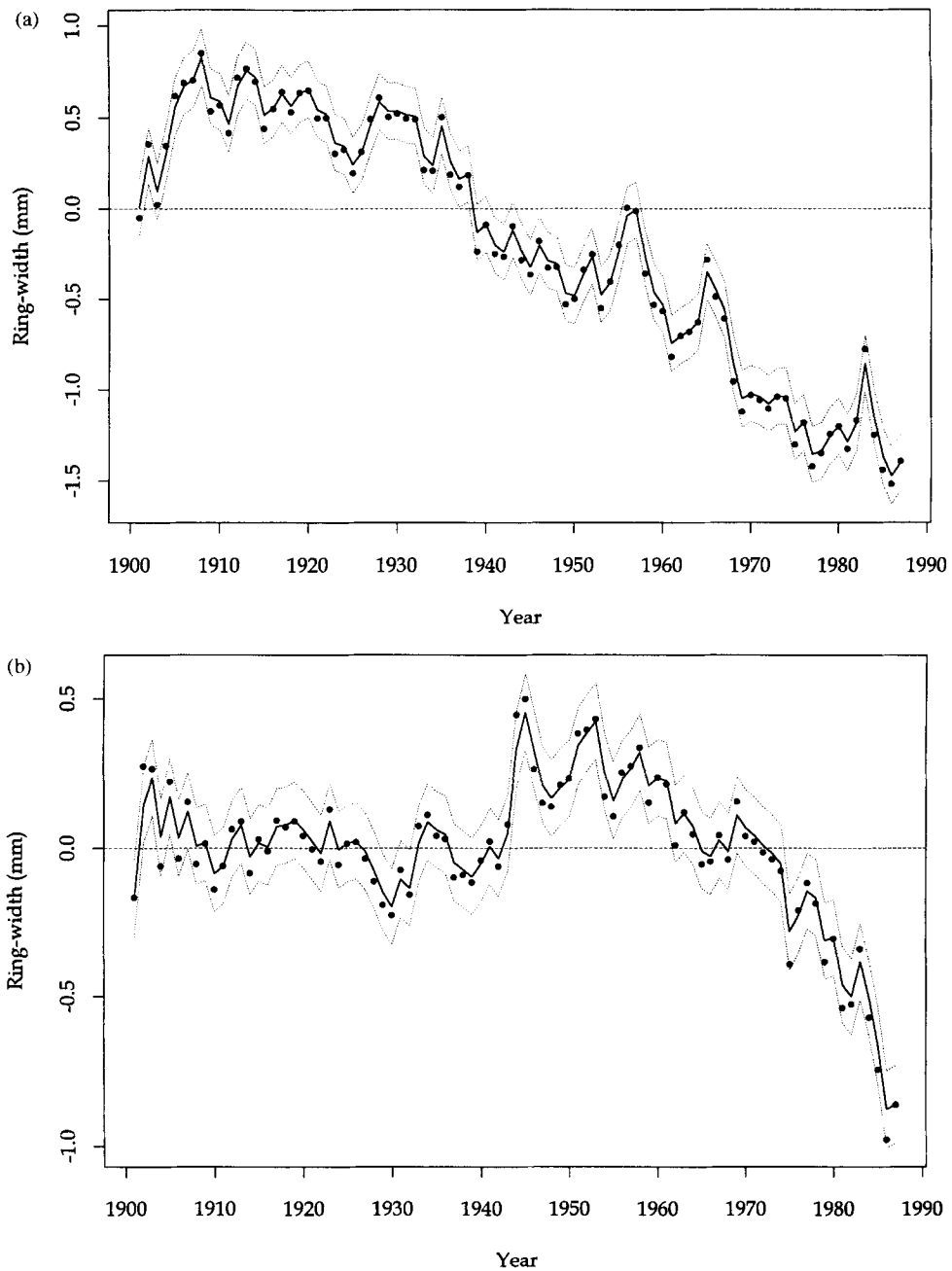


Figure 2a–b. Susceptible minus resistant tree-ring chronology (see text for explanation) y_t , (•); optimal Kalman filter estimates (solid line) and 95% confidence intervals for y_t (dashed lines) for the (a) codominant crown class ($\max(n_t) = 9$ pairs), and (b) intermediate crown class ($\max(n_t) = 16$ pairs) on the Downeast site.

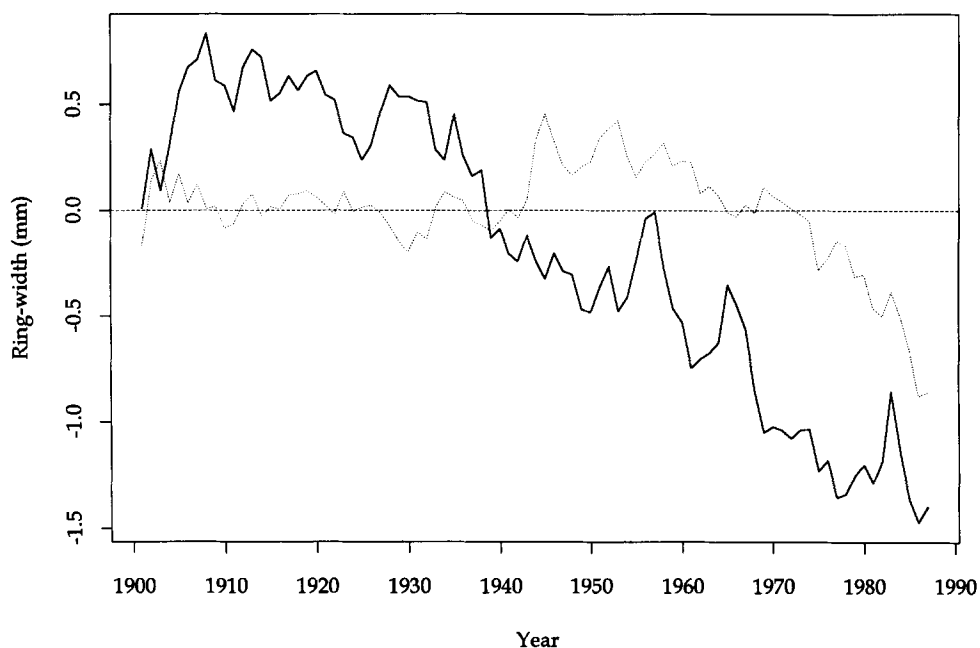


Figure 3. Optimal Kalman filter estimates for the codominant (solid line) and intermediate (dashed line) crown classes on the Downeast site.

significant decline in 1962. As on the Downeast site, the minimum growth difference was estimated to occur in 1986 with $a_t = -1.12$ mm.

The time-variant RWP model also fitted the intermediate crown class the best with average $k_t = 0.817$. Growth of susceptible trees was better than that of resistant trees until 1963, was comparable for the next six years, and then sharply declined thereafter (Fig. 4b). There is no good explanation for the three years (1977–78 and 1982) of relative increase that occurred after this time. The minimum growth difference was estimated at $a_t = -0.90$ mm in 1990.

The link to stand dynamics is not as clear in the Northern trees as in the Downeast sample, although some interaction between crown classes is apparent (Fig. 5). Until the early 1930s, the relative growth of the susceptible codominants was somewhat greater than that of the intermediate crown class. However, as growth of the susceptible and resistant codominants became more equitable (mid 1930s to the early 1960s), there was a slight increase in growth of the susceptible intermediates (from the late 1930s to the late 1950s) relative to their resistant partners, though probably not statistically significant. The final decline of the susceptible trees was within six years of each other for both crown classes, again suggesting a reduced influence of crown stand dynamics as compared to the Downeast site.

4. Discussion

Analysis of the annual ring measurements revealed a substantial difference in the onset of the growth declines for susceptible trees between the two regions. For the codominant crown class of full vigour, the Northern site decline lagged behind the Downeast decline by 21 years. This may reflect the spatial–

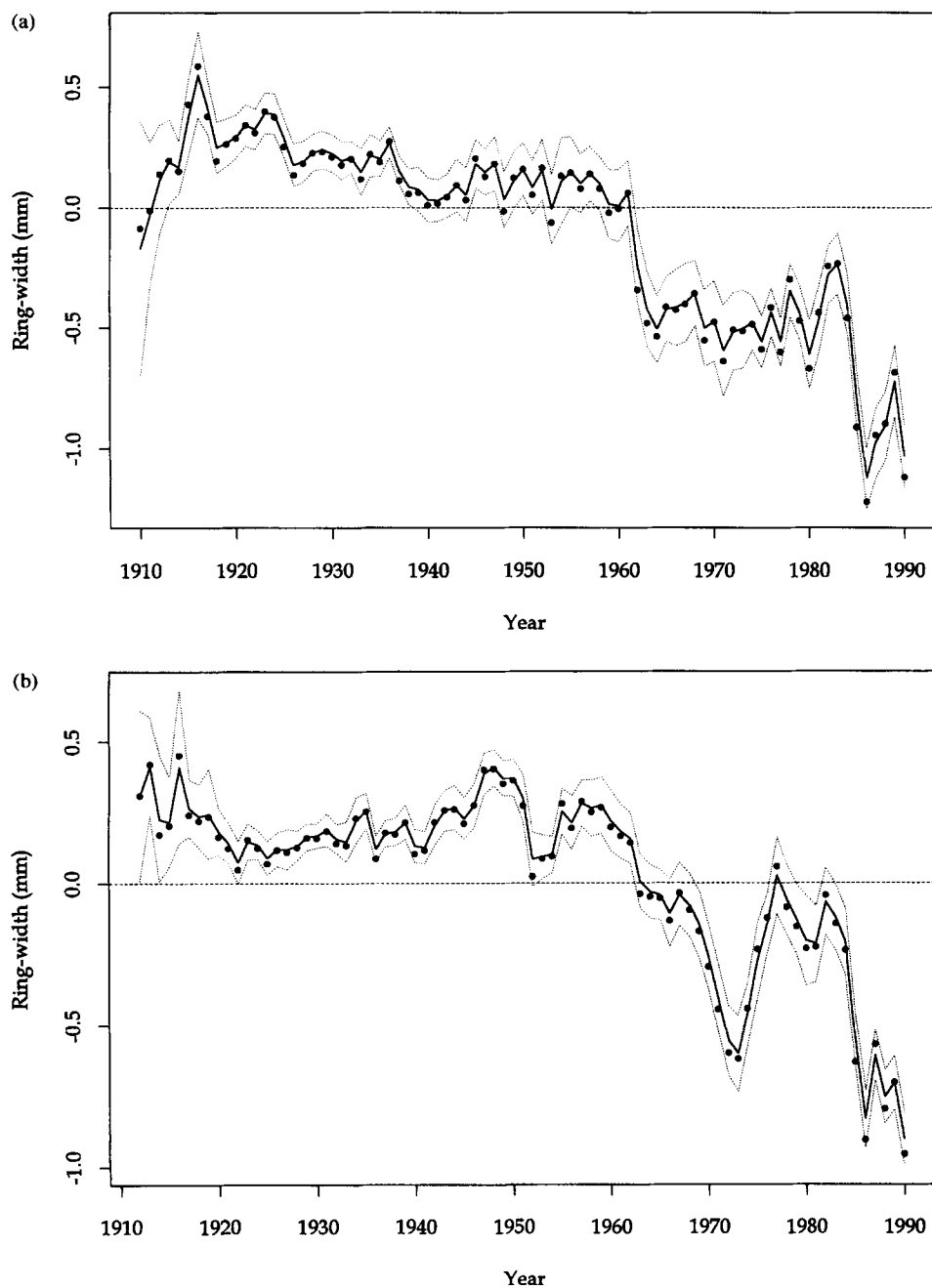


Figure 4a–b. Susceptible minus resistant tree-ring chronology (see text for explanation) y_t , (•); optimal Kalman filter estimates (solid line) and 95% confidence intervals for y_t (dashed lines) for the (a) codominant crown class ($\max(n_i) = 34$ pairs), and (b) intermediate crown class ($\max(n_i) = 32$ pairs) on the Northern site.

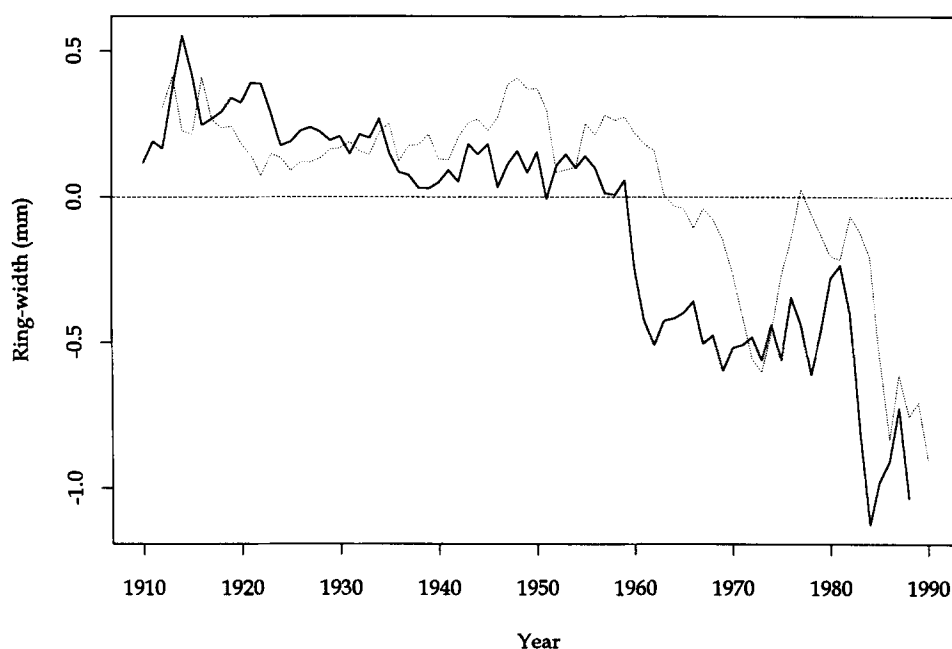


Figure 5. Optimal Kalman filter estimates for the codominant (solid line) and intermediate (dashed line) crown classes on the Northern site.

temporal patterns of spread of the disease agents that probably occurred throughout the region (Houston, 1994). However, growth decline differences could also be attributable, at least in part, to the differences in climate between the two sites. In general, the Downeast site is characterized by a more coastally-moderated climate, 22–35 km north of the Gulf of Maine, while the Northern site, being 106 km north of the coast, is far from such influences.

Houston and Valentine (1988) corroborated results of other studies conducted in Europe with their findings that heavy October rainfall coupled with severely cold winters were negatively related to the amount of stem surface cankered by the BBD complex in the following growing season. They proposed that these factors contributed to a reduction in the scale insect population numbers; whether they also affected the other components of the BBD complex is, as yet, unknown. Thus, environmental variables may play a key role in the different growth decline patterns of the two sites. The coastally-moderated climatic environment of the Downeast site is likely to have been more conducive to rapid, large build-ups in scale populations. If these environmental conditions also favoured the subsequent infection by the *Nectria* fungi, then the sharply delineated patterns of decline with respect to associated stand dynamics discussed above seem logical. With few environmental factors limiting disease development, the influence of stand dynamics (availability of susceptible beech intermediates following the death and/or decline of larger, more susceptible codominant trees) becomes more clearly discernible.

In contrast, beech trees on the ridge-top of the Northern site are regularly exposed to severely cold winter temperatures (-34°C) which have been found to be lethal to *C. fagisuga* (Barter, 1953), or associated with declines in their populations (Houston and Valentine, 1988). In response to these harsh conditions, it is probable that *C. fagisuga* populations did not increase as rapidly or remain at high levels as long as did those on the more moderate sites. One might imagine (in the absence of

populations censuses from this area) that scale populations slowly increased over time in a 'saw-toothed' pattern (as did the 'population' of beech bark disease stem cankers in similarly northern sites (Houston and Valentine, 1988)), with high winter mortality always contributing to a decline in the numbers of *C. fagisuga*, but fecundity of the survivors always enough to continue the overall upward population trend (Pielou, 1974, p. 3). Following this scenario, the effects of environment and stand dynamics must interact on this site. The susceptible codominant trees slowly declined with respect to the resistant trees as the beech scale population slowly built over time. As the susceptible codominants slowly weakened, with their crowns eventually becoming progressively thinned from the effects of the disease, the somewhat less susceptible intermediate trees slowly responded with increased growth. All the while, cankers resulting from *Nectria* infection accumulated and continued to weaken trees until the final decline period. Under this scenario, the final, rather sharp relative growth decline in susceptible trees may be due either to the long-term cumulative effects of the repeated cankering, or to a quick, sustained build-up of the scale population, and subsequent heavy *Nectria* infection, triggered by the occurrence of a series of mild winters, or both. In addition, environmental conditions may have been favourable to the growth of trees not impacted severely by the disease.

Gavin and Peart (1993) noted the long-term persistence of beech infected by BBD in the aftermath forests, with survivors continuing since infection in 1949 on one of their sites. Our data confirm this with trees on the Downeast site, probably having been infected in the early 1930s and declining significantly by 1941, still living by the time the data for this study were collected after the end of the growing season in 1987.

Finally, our results conform well to those of the proposed pattern of spread of the BBD complex throughout the north-east region of North America as given by Houston *et al.* (1979) and Houston (1994). Dates for *discovery* of the beech scale given by these authors place the insect in the Downeast area by at least 1935 and in the Northern area circa 1935–40. Their isopleths of insect occurrence are only approximate, and our results in the Northern area might suggest a slightly earlier arrival.

5. Concluding remarks

The Kalman filter, with its ability to model non-stationary time series like the tree-ring series presented here, appears very useful for trend monitoring in general. However, this study would be incomplete if some important results found in the process of fitting the hyperparameters with MLE were not mentioned. The estimation process in fitting these hyperparameters is often not straightforward, especially for the more complicated models, and requires some degree of skill and scrutiny.

The reason that both simplex and scoring were used in our computer program is that often, in the case of the LLT model, the brute-force simplex method is needed to arrive in the neighbourhood of the optimum before the subsequent employment of scoring from that simplex stopping point to converge on the optimum. Occasionally, simplex appeared to be close to a local optimum such that ψ_β was on the order of 10^{-4} and the associated score vector component on the order of $|10^{-2}|$, with all other score components effectively zero; however, scoring would then subsequently diverge from the simplex point. These latter cases were normally associated with points near the boundary of the admissible region for ψ_β . In such cases, $\psi_\beta \rightarrow 0$ and the final estimates for ψ_y and ψ_μ were very close to those of the RWPN model. In the case of the LLT model, ψ_β fell on the boundary of the parameter space ($\psi_\beta = 0$) in about 60 per cent of the models fitted. The simpler RWPN model, however, always converged with the scoring algorithm in relatively few iterations and never required simplex to find an initial starting point. Harvey and Peters (1990) noted very

similar behaviour for the hyperparameter associated with the slope in their LLT model when using time domain MLE procedures in an analysis of six macroeconomic time series.

Even though the RWP model always converged easily, occasionally $\mathbf{Q}_t/h_t$ was large yielding a ψ_y estimate on the order of 10^{-2} . As mentioned earlier, this situation is one of all data and no model. In this case the optimal update variance estimate, $\mathbf{P}_t$, also becomes very small and confidence intervals associated with μ_t are essentially useless. These results indicated that either the RWP model itself or the variance model (time-variant or time-invariant) was inappropriate and an alternative structural or variance model was chosen. Indeed, with some categorizations of the pairs other than codominant–intermediate reported here, the LLT model appeared to be the more appropriate model because of these problems.

Useful guidelines pertaining to the minimum length of a time series (T) that may be used to estimate the hyperparameters with ML seems to be lacking in the literature. In addition, nothing has been reported in the dendrochronology literature on the minimum sample size n_t needed by year to estimate $\sigma_{y_t}^2$ and $\sigma_{\beta_t}^2$, and the interplay between these variance components and estimation of the hyperparameters. For example, when the $n_t = 2$, pairs from one of the plots in the Downeast group were used with a series length of $T = 31$ years, the Kalman filter estimates converged well; however, the optimal estimates $\hat{a}_t$ resembled those of a smoothed estimator that had filtered out all of the high frequency signal. In another case with $T = 37$ and $n_t = 6$, the effects were similar. In other cases where n_t varied from two to six pairs, but the series was longer ($T \approx 90$ years), the estimates showed no propensity towards a low-frequency smoother. In conjunction with their study of macroeconomic time series, Harvey and Peters (1990) offer only that the likelihood surfaces are relatively flat for the short (37 observations) series considered. Clearly, there is a need for more study on the dependence of the variance and structural model used on sample size (and associated estimates of $\sigma_{y_t}^2$ and $\sigma_{\beta_t}^2$) and series length for tree-ring analysis.

Finally, several of the previous papers utilizing the Kalman filter in dendrochronological or dendroclimatological studies have employed the smoothing equations associated with the Kalman filter for optimal smoothed state vector estimates (Cook and Johnson, 1989; Van Deusen, 1991a; Visser and Molenaar, 1988). We did not employ smoothing because of the tendency of the method to obscure the details of the individual yearly variations that were sought in this study to determine the growth dynamics associated with the disease complex. Gove (1996) gives examples and discusses this in more detail and describes computer programmes written in the GAUSSTM matrix language that are available for analyses similar to those presented in this study.

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